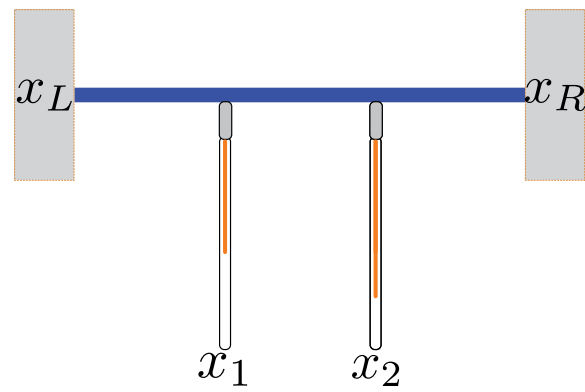






Recall the inhomogeneous system modelling the insulated metal rod with the temperatures at its two ends fixed at  $\boldsymbol{x}_L$  and  $\boldsymbol{x}_R$ , and with two thermometers placed at equal distance along it:

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} x_L \\ x_R \end{pmatrix}$$

Homogeneous solution

Let us first consider the case where the end points are held fixed at **0** degree Celsius, so  $\boldsymbol{x}_L = \boldsymbol{x}_R = \mathbf{0}$ . This makes the system **homogeneous**.

The eigenvalues of the system are  $\lambda_1 = -1$  and  $\lambda_2 = -3$ . The corresponding eigenvectors are  $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$  and  $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ .

Hence, a fundamental matrix of the homogeneous system is given by the product:

$$\mathbf{X}(t) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-3t} \end{pmatrix}.$$

Review: phase portraits

1/1 point (graded)

As above, consider the homogeneous system describing the rates of change of temperatures  $\boldsymbol{x}_1$  and  $\boldsymbol{x}_2$  along a metal bar whose two ends are fixed at **0** degree:

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

What is the phase portrait of this system?

☐ Stable spiral

☒ Stable node

☐ Saddle

☐ Unstable spiral

☐ Unstable node

☐ None of the above



Solution:

Since the eigenvalues are distinct, negative, and real, the phase portrait is a stable node. In particular, the homogenous system exhibits no oscillations, and both temperatures  $\boldsymbol{x}_1$ ,  $\boldsymbol{x}_2$  will tend to zero over time, as expected for a rod held at 0 degrees at both ends.

## Variation of parameters practice

4/4 points (graded)

We now consider the case in which the left end of the rod is held at 0 degrees, so  $x_L = 0$ , but the right end is held at 100 degrees, so  $x_R = 100$ . The inhomogeneous system is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r} \quad \text{where } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 0 \\ 100 \end{pmatrix}.$$

Variation of parameters says that the general inhomogeneous solution is  $\mathbf{X}\mathbf{v}$ , where  $\mathbf{v} = \int \mathbf{X}^{-1}\mathbf{r}dt$ , and  $\mathbf{X}$  is a fundamental matrix of the associated homogeneous system. In this problem, we will carry out the necessary computations.

Compute  $\mathbf{X}^{-1}\mathbf{r}$  using the fundamental matrix we previously found,

$$\mathbf{X}(t) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-3t} \end{pmatrix}.$$

(Enter a single vector as your answer. Enter  $[\mathbf{a}, \mathbf{c}]$  for the vector  $\begin{pmatrix} a \\ c \end{pmatrix}$ . That is, use **comma** to separate the entries, and **square bracket** the entire vector. )

$\mathbf{X}^{-1}\mathbf{r} =$   ✓ Answer: [50\*e^t, -50\*e(3\*t)]

Integrate each component to find  $\mathbf{v} = \int \mathbf{X}^{-1}\mathbf{r}dt$ . Use  $c_1$  and  $c_2$  as the constants of integration for the first and second components respectively.

(Type  $c\_1$  for  $c_1$  and  $c\_2$  for  $c_2$ . You must enter the simplest possible integration constant. For example  $5c_1$  should be entered as  $c_1$ , similarly  $c_1 + 1$  should be entered as  $c_1$ .)

$\mathbf{v} = \int \mathbf{X}^{-1}\mathbf{r} dt =$   ✓ Answer: [50\*e^t+c\_1, -50/3\*e(3\*t)+c\_2]

Find the general inhomogeneous solution by computing  $\mathbf{X}\mathbf{v}$ , with the fundamental matrix  $\mathbf{X}$  given above, and the  $\mathbf{v}$  you just computed. Enter the two components  $x_1(t)$  and  $x_2(t)$  separately below. (Your answer should involve the constants of integration  $c_1$  and  $c_2$  from the previous step.)

$x_1(t) =$   ✓ Answer: c\_1\*e^(-t)+ c\_2 \*e^(-3\*t) + 100/3

$\frac{100}{3} + c_1 \cdot e^{-t} + c_2 \cdot e^{-3 \cdot t}$

$x_2(t) =$   ✓ Answer: c\_1\*e^(-t)- c\_2\*e^(-3\*t) + 200/3

$\frac{200}{3} + c_1 \cdot e^{-t} - c_2 \cdot e^{-3 \cdot t}$

? FORMULA INPUT HELP

**Solution:**

$$\begin{aligned} \mathbf{X}^{-1}\mathbf{r} &= \frac{1}{2} \begin{pmatrix} e^t & 0 \\ 0 & e^{3t} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 100 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} e^t & 0 \\ 0 & e^{3t} \end{pmatrix} \begin{pmatrix} 100 \\ -100 \end{pmatrix} \\ &= \begin{pmatrix} 50e^t \\ -50e^{3t} \end{pmatrix} \end{aligned}$$

Integrating component wise, we get

$$\int \mathbf{X}^{-1}\mathbf{r} dt = \begin{pmatrix} 50e^t + c_1 \\ -50e^{3t} + c_2 \end{pmatrix}$$

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## Accessibility

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